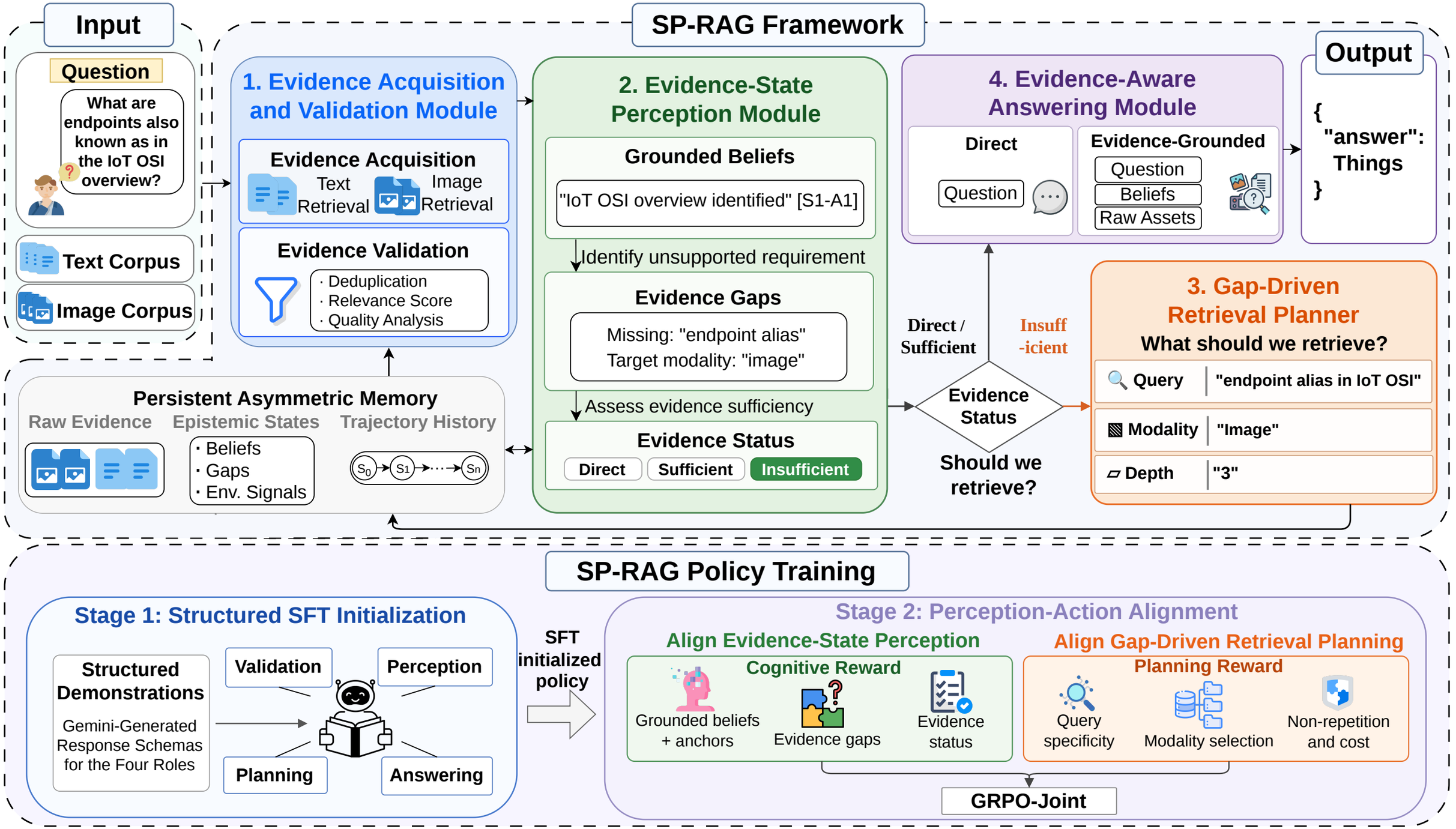

Input
Question
What are endpoints also known as in the IoT OSI overview?
Text Corpus
Image Corpus
SP-RAG Framework
1. Evidence Acquisition and Validation Module
Evidence Acquisition
Text Retrieval
Image Retrieval
Evidence Validation
· Deduplication
· Relevance Score
· Quality Analysis
2. Evidence-State Perception Module
Grounded Beliefs
"IoT OSI overview identified" [S1-A1]
Identify unsupported requirement
Evidence Gaps
Missing: "endpoint alias"
Target modality: "image"
Assess evidence sufficiency
Evidence Status
Direct
Sufficient
Insufficient
4. Evidence-Aware Answering Module
Direct
Question
Evidence-Grounded
Question
Beliefs
Raw Assets
Output
{
"answer": Things
}
Direct / Sufficient
Insuff -icient
Evidence Status
Should we retrieve?
3. Gap-Driven Retrieval Planner
What should we retrieve?
Query
"endpoint alias in IoT OSI"
Modality
"Image"
Depth
"3"
Persistent Asymmetric Memory
Raw Evidence
Epistemic States
· Beliefs
· Gaps
· Env. Signals
Trajectory History
S0 S1 ··· Sn
SP-RAG Policy Training
Stage 1: Structured SFT Initialization
Structured Demonstrations
Gemini-Generated Response Schemas for the Four Roles
Validation
Perception
Planning
Answering
SFT initialized policy
Stage 2: Perception-Action Alignment
Align Evidence-State Perception
Cognitive Reward
Grounded beliefs + anchors
Evidence gaps
Evidence status
Align Gap-Driven Retrieval Planning
Planning Reward
Query specificity
Modality selection
Non-repetition and cost
GRPO-Joint